

Construction Element Detection for Site Progress Verification

A cloud-only YOLOv8 pipeline for automated progress evidence

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EXECUTIVE SUMMARY

Site progress in the Gulf is still recorded by photograph and narrative, with a two- to seven-day gap between capture and record. The photograph is archived; the information inside it is not. ISO 19650 asks for information delivered into a common data environment in a structured form, and a JPEG in a folder does not meet that bar.

This project trains a YOLOv8 detector on five construction classes — **steelbar, brick, pvcpipe, scaffold and excavator** — so that a site photograph yields a structured detection record: image, time, zone, class, confidence, bounding box. Steelbar, brick and pvcpipe act as proxies for structural, masonry and MEP first-fix activity in a zone; scaffold and excavator report site condition — temporary works standing, plant active.

The system is deliberately scoped as a **triage tool**. It directs a person to the photographs worth examining. It does not replace the examination, and it has no safety function of any kind. The entire pipeline runs in the browser — Roboflow for data, Google Colab for training — with no local installation, so a third party can reproduce every result from the repository alone.

METHOD

A public dataset, *seungyeon/construction-site-km7bh* (700 images, CC BY 4.0), was forked into our own Roboflow workspace, re-split from 70/20/10 to the required 80/20, and extended with our own images — including deliberate boundary cases: the objects that resemble a class and are not it. Loose scaffold tube against steelbar, a ladder against scaffold, a wheeled loader against excavator.

Every label follows a written class-definition contract that specifies inclusions, exclusions, the one-box-per-visually-separable-group rule and a 20-pixel minimum size. The contract exists because without written rules, annotation drifts between people and the model learns the drift rather than the object. SAM-assisted labelling was evaluated: it accelerated geometry on isolated objects and failed on grouped and amorphous classes, because it automates the drawing and never the judgement.

Dataset images	700
Split	560 / 140 (80 / 20)
Model	yolov8s.pt
Epochs	30
Image size	640
Batch	16
Hardware	Tesla T4, 15360 MiB
Training time	6.4 min

RESULTS

Class	Precision	Recall	mAP@50	mAP@50-95
All classes	0.950	0.884	0.943	0.809
brick	0.984	1.000	0.995	0.932
excavator	0.969	1.000	0.995	0.883
pvcpipe	0.986	0.930	0.964	0.835
scaffold	0.955	0.875	0.979	0.858
steelbar	0.855	0.612	0.781	0.535

INTERPRETATION

1. **All three success criteria were met, and that is the least interesting result.** mAP@50 reached 0.943 against a 0.50 target. Clearing a bar that far is itself a signal that the measurement is wrong, not that the model is excellent.
2. **The split leaks.** Consecutively numbered frames sit on both sides of it (steelBar_5397k in train, 5398k in validation); mAP@50 was already 0.677 after one epoch; two classes reach recall 1.000. The headline number is not a generalisation estimate.
3. **steelbar is weakest in both architectures tried** — 0.781 here, 0.539 in an independent YOLOv11n cross-check — despite having the most training instances. Shape and our group-boxing rule explain it; instance count does not. Per-class ranking survives the leak; absolute values do not.

ERROR ANALYSIS AND NEXT ITERATION

Three false positives and three false negatives were examined against ground truth and documented with filenames and hypotheses in *docs/error_analysis.md*. Each hypothesis is tied to a cause that the data can address — an under-represented class, an object scale the source resolution cannot support, or a class boundary left ambiguous in the definition contract — rather than to model capacity. Three prioritised dataset improvements follow from them, each naming a concrete action and the specific metric it should move.

RISK: THE ASYMMETRY BETWEEN THE TWO ERRORS

A false negative is silent. Steelbar is placed, the model misses it, no record is created, and nobody investigates because nobody knows there is anything to investigate. The gap propagates into the programme unchallenged. **A false positive is loud.** A phantom record is contradicted by the next site walk. It costs a wasted verification trip and is self-correcting.

For progress logging, therefore, **recall is the priority** and the operating threshold should be set low. For site-condition alerts the asymmetry reverses: repeated false alarms teach supervisors to ignore the system, and an ignored alert is worse than no alert. **One model, two thresholds** — low for logging, high for alerting — both read off the precision-recall curve rather than left at the default.

LIMITATIONS

- The reported metrics are inflated by train/validation leakage from near-duplicate frames. They measure memorisation as much as detection. Re-splitting by group is improvement D1.
- No safety function. The model detects no people, no PPE and no hazards. Treating it as a safety system would reduce site safety, not improve it.
- Presence only. It cannot measure quantity, spacing, cover or conformance, and its detection counts must never enter a payment application.
- Validated scope is daytime site photography at working range. Night, dust, rain, drone and enclosed-space imagery are untested.
- Human in the loop is mandatory. Every output is a prompt to look, not a decision.

GOVERNANCE AND LICENSING

Artefact	Licence	Owned	Constraint
Code and notebooks	MIT	Yes	None
Base dataset	CC BY 4.0	No	Attribution; referenced, not redistributed
Images we added	CC BY 4.0	Yes	Attribution
Trained weights	AGPL-3.0	Derived	Commercial use needs AGPL compliance or a licence

No class in this model detects, identifies or tracks people — a design decision, not an accident. Adding a person or PPE class would move the system into the EU AI Act's high-risk employment category and trigger conformity assessment, logging and human-oversight obligations. The boundary is one class label away.